

Machine Learning Based Crop and Fertilizer Recommendation for Sustainable Agriculture in Sri Lanka

Web Based Crop and Fertilizer Recommendation
Application

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A modern, bright interior space, possibly a lounge or office, with large windows, indoor plants, and a teal banner. The banner contains the text "Problem Statement".

Problem Statement

Many Sri Lankan farmers rely on traditional practices and personal experience when choosing crops and fertilizers. This often leads to

- **poor yields**
- **soil damage**
- **inefficient resource use.**

There is a lack of accessible, data driven tools to guide agricultural decisions. This project addresses that gap by introducing a machine learning based recommendation system tailored to local conditions.



Objectives

01



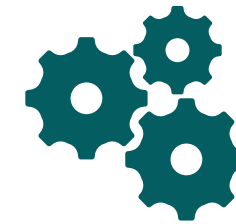
To develop a machine learning based crop recommendation system that considers various environmental and soil parameters.

02



To design and implement a fertilizer recommendation model based on soil nutrients, crop type, and environmental factors.

03



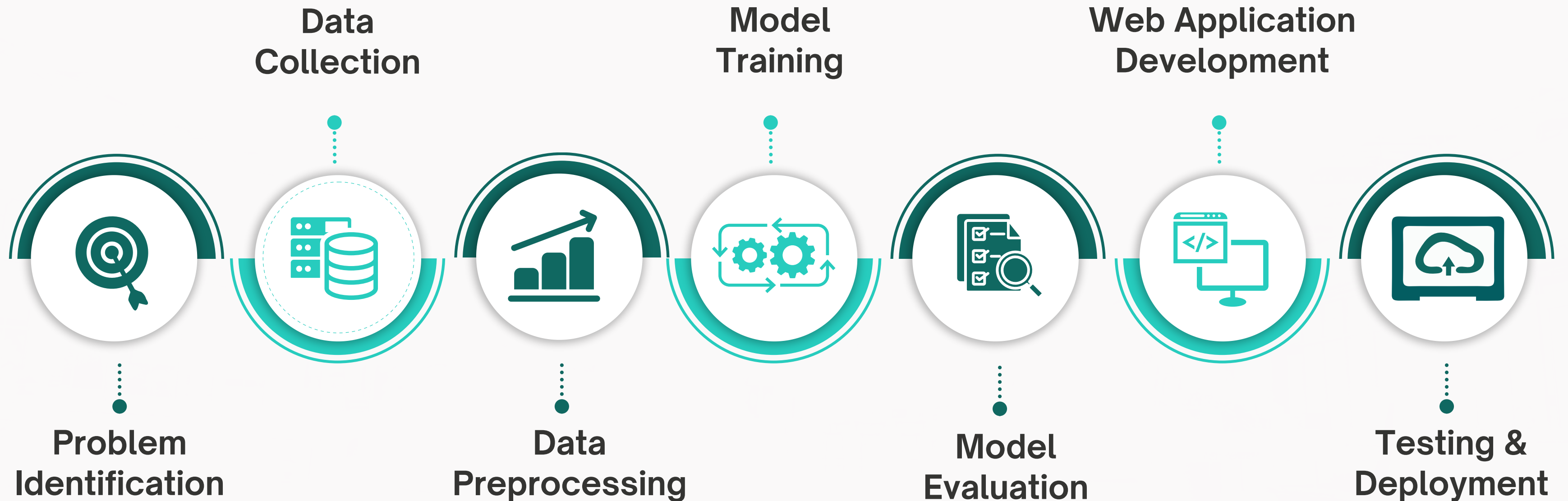
To evaluate the performance of the machine learning models using appropriate evaluation metrics.

04



To integrate both crop and fertilizer recommendation models into a single user friendly web application

Methodology



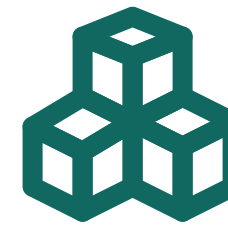
Key Findings

Accuracy



The machine learning models for both crop and fertilizer recommendation showed high accuracy, with the crop model achieving over 90% accuracy.

Preprocessing



Preprocessing techniques like label encoding and oversampling helped address imbalanced data effectively.

Integration



The hosted web based application integrated both models successfully, providing real-time, user-friendly predictions.

Impact



The system shows strong potential to support informed agricultural decisions, especially in data-scarce regions like Sri Lanka.

Novelty



Localized Focus

- Designed specifically for Sri Lankan agriculture, adapting non-local datasets effectively.



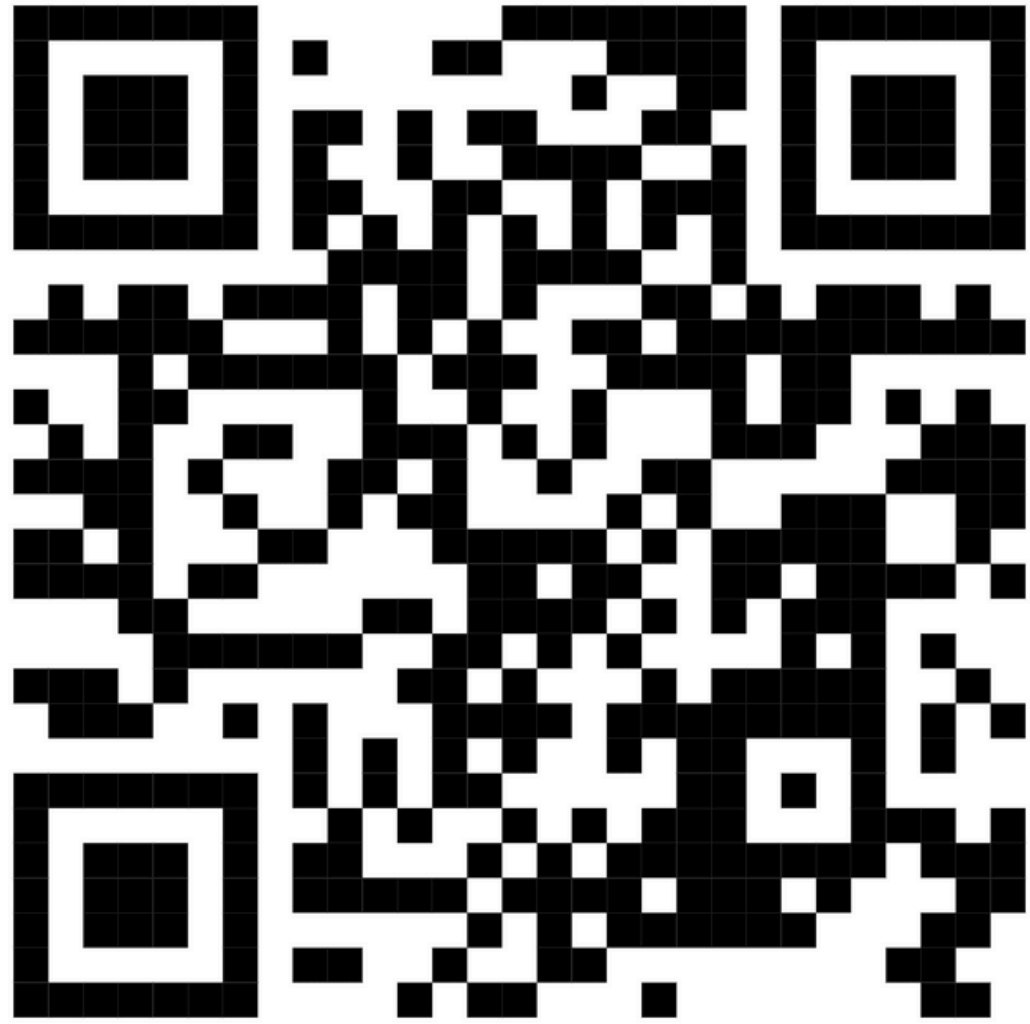
Fully Deployed System

- A complete ML-powered web application that is hosted and accessible online.



User-Friendly Design

- Provides real-time, web-based recommendations with minimal input needed from the user.



THANK YOU

For watching this presentation

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